

Manual

Staff-Related BDE Functions BDE-PBF 8.2

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1 Overview: Staff-Related BDE Functions

Purpose

The function package provides functions for collection and posting of order-related labor utilization.

Implementation notes

Use the function package if:

- You wish to enter order-related labor utilization
- You wish to employ comparison mechanisms between the Personnel Time Management (PZW) module and the Shop Floor Data Collection (BDE) module with respect to the entered labor utilization,
- You require specific identifiers with respect to the labor utilization for wage payment reasons,
- You wish to employ personalized plausibility checks for the entry/posting of personnel and orders.

Integration

You can evaluate the recorded or entered data in MOC using the applications of the [BDE-PMA](#) function package. If corresponding interfaces are used, data can also be uploaded to a higher-level ERP system.

The HR master data required to carry out postings are either transferred from a higher-level system or are edited directly in HYDRA.

Features

- Operator positions
 - Configuration and entry of a workplace-specific operator position when a person logs on
- Wage/premium indicator
 - Configuration and entry of a wage/premium indicator when a person logs on.
- Personnel logon/logoff
 - Logons and logoffs of one or more persons to and from operations or the workplace.
- Calculation of staff-related order times
 - Calculation of staff-related order times as the basis for performance-related wage calculation. Entry of the labor utilization per operation.
- Posting rules
 - Different staff-related posting at individual workplaces and group workplaces.
- Shift model comparison
 - Comparison of recorded labor times with the BDE shift model stored at the workplace or optionally with the person's individual BDE shift model.

- Multi-machine operation
 - Splitting of the order-related labor times for multi-machine operation at individual workplaces.
- PZE controls BDE
 - Possibility of logging off staff logged on to operations when clocking out and optional logon when clocking in the next time.
- Waiting period processing
 - Waiting time monitoring and posting of these times without logon to waiting period operations.
 - Rounding of the first order logon by analogy with the clock-in of the employee.

2 Operator Position

Overview

Menu	Master data => Workplaces/Machines => Operator Positions
Transaction code	oppo
Function authorization	mdoppo

Purpose

The employee enters the operator position (= function) when logging on to a workplace/machine. They select from a list of predefined operator positions/functions. Examples of operator positions/functions are: machine operator, assistant, setter,

Requirements

The workplace has to be created.



If you want to be able to select or enter an operator function when logging on to the system, the settings in the entry client needs to be customized.



Configuration changes

In order for the settings or changes made to be interpreted by the AIP, the AIP that is assigned to the workplace/machine must be restarted.

Selection criteria

The application provides the following selection criteria:

Workplace

Selects the workplace

Short name

Short name of the workplace.

Group

Group of the workplace.

Cost center

Cost center of the workplace.

Field descriptions

Operator position/function

The operator position has been designed as posting criterion for the entry.

Short name

Abbreviation (not displayed during entry)

Description

Description of the operator position to be displayed during entry

Function

Control function

Order type

Reserved Currently not processed Please leave field empty

Maximum number of people

By entering a value greater than 0, the maximum number of people allowed to log on with this operator position, may be limited.

3 Wage and Premium Indicators

Summary

Menu	Master data → Workplaces / machines → Wage indicators
Transaction code	Prin
Function authorization	Mdprin

Usage

Premium indicator is a key term that refers to a wage payment. Each employee is assigned to a certain premium based on which category he or she falls under according to the relevant collective bargaining agreement. If his or her activity in production deviates from plan specifications then the BDE system will record this. The employee provides his/her premium indicator when logging on to a machine. HYDRA transmits the premium indicator to the payroll accounting system. There, the wage to be paid to the employee is calculated based on a target/ actual comparison.



This function is only available at the Windows based CTWIN or AIP terminals.

Selection criteria

The application provides the following selection criteria:

Workplace

Unique workplace number for which premium indicators are recorded.

Field descriptions

Workplace

Workplace number

Premium indicator

Premium indicator (ID). The worker must enter this ID at the shop floor terminal.

Designation

Description of the premium indicator

4 Waiting Period - Person

Utilization

Waiting period processing is an optional function, which allows for complete recording of labor times in shop floor data collection (BDE). This applies to both, gaps between individual personnel postings in BDE as well as between the personnel postings in BDE and the clock-in or clock-out postings of the time & attendance module (PZE).

Even if time & attendance functions are not integrated, a very good posting quality can be achieved in BDE (shop floor data collection) if the advanced logon time function, waiting period processing and appropriate options to log off staff automatically at the end of shifts are in use. This can still be improved for flexible working times if the option "PZE (in/out) controls BDE" is used.

Integration

Provided that the time & attendance component (PZE) is in use, the waiting period processing function can also be controlled by PZE clock-in and clock-out records.

Prerequisite

The waiting period processing is enabled in the basic parameter settings --> BDE --> Waiting period processing BDE --> Activate waiting period processing.

The function can be used in connection with the time & attendance module (PZE). However, using PZE is not mandatory. If integration with PZE is required, the relevant options needs to be enabled in the basic parameter settings --> BDE --> PZE (in/out) controls BDE".

Configuration options

The personal waiting period processing is configured across the system in the basic parameter settings. The document dealing with the basic parameter settings provides further information on this.

Filling the gaps between BDE personnel postings

The "waiting period" is defined in minutes within the basic parameter settings if waiting period processing is enabled. It indicates the maximum time that may pass between a logoff at an operation and the logon to the next operation or it indicates within what time an employee has to log on to an operation after his/her clocking-in. If this waiting period is not exceeded, the person's logon is dated back to the time he/she logged off at last or to his/her clocking-in. In case the waiting period is exceeded, the system posts the time exceeded to the defined waiting period operation.

A "maximum waiting period" is also defined within the "basic parameter settings". The maximum waiting period indicates the period of time as of which HYDRA rejects waiting period postings. If the maximum waiting period is set to 0, waiting period postings will not be rejected. The system always generates a waiting period posting if waiting period processing is enabled. Even at the end of a working day, it starts with the time when an employee logs off last and ends with the first logon on the next day, provided it is not interrupted by clocking-out if the control by PZE option is enabled. This data record spanning two working days of an employee can be suppressed by the maximum waiting period.

The waiting period has to be posted to a "resource performance account (RPA)". An account has to be defined within the basic parameter settings for posting purposes. Please note that only the account number may be entered, e.g. 9 for RPA U8 or 10 for RPA U9. These accounts are free accounts and can be used for waiting period processing without affecting the events.

Options with "PZE (in/out) controls BDE"



The functions of "PZE controls BDE" process a BREAK clocking (beginning of break) as a clocking-out. The end of the break corresponds to a clocking-in.

No control

Clocking records pertaining to the time & attendance module (PZE) do not result in employees being logged on or off in BDE and do not cause any waiting period processing.

Waiting period processing

Processing is enabled if the waiting period is activated in BDE --> PZE (IN/OUT) controls BDE within the basic parameter settings.

When clocking-in, the person is automatically logged on to a waiting period operation in the shop floor system.

If the user logs on to a production OP during the "waiting period" the duration between the clocking-in and the logon to the production operation is posted onto the registered operation. The logon of the production operation is posted forward to the posting in HYDRA-PZE.

The system generates a "waiting period posting" as soon as the waiting period is exceeded. The data posted to the production order is not changed. This processing is also activated for the times between the logoff and logon of staff during the working time.

Once the person clocks out in the time & attendance module, they are automatically logged off in HYDRA-BDE as well. Provided that the person was no longer logged on to an operation at the time of clocking out, a waiting period posting is generated for the period of time between the last logoff of the person and the clocking-out. This is even the case if the waiting period is not exceeded.

The operation is interrupted automatically if the last person who works on the operation clocks out.

Waiting period processing with "auto in"

Waiting period processing is activated, provided that it is enabled with "auto in" in the basic parameter settings --> BDE --> PZE (IN/OUT) controls BDE. In this case, however, the person is automatically logged on to the operation that was posted the last time by this person.

As a prerequisite for this logon, this operation must neither be active nor finished already.

In case the operation could be logged on, the person is also logged on to the operation. In this case, however, the person is always directly logged on; there is never an advance logon.

The logon is made on that workplace to which the person and the operation were logged on before clocking out. If it is an individual workplace all people who are already active are connected to the new operation and the new person is connected with all OPs that are already active at this workplace.

If an advance logon of staff is required instead, it might be configured by an entry in the INI configuration:

Parameter name	Parameter value
Name	PZE_CONTROLS_BDE
Key	ADVANCED_LOGON_OF_STAFF
Value	TRUE / FALSE
Comment	Generate advance logon of staff optionally

If this processing is activated by the INI entry

- An advance logon of staff is generated instead, when clocking in, provided that the person clocks in within the advance logon period for the machine or the machine is in the "no shift" status (time without shift).
- No OP is logged on, if the person clocks in at a time when the machine is in the "no shift" status and the operation to be posted is in the "automatically interrupted" status.

Waiting period processing with clock-out

If the waiting period processing with clock-out is activated in the basic parameter settings --> BDE --> PZE (IN/OUT) controls BDE, the person is automatically logged off from all machines and orders as soon as they clock out. In addition, the operation is also interrupted, if the last person who is logged on to the operation is logged off.

General notes



Operations are not logged on automatically by the function "PZE controls BDE", provided that resources are posted for the operation or inspection operations are connected with the operation.

In such cases, only the person is logged on or off by the PZE clocking.

When it comes to group workplaces or overhead cost operations, it might also be the case that the person cannot be logged off automatically in the cases mentioned above, as one person always has to be logged on to an operation.

When operations are merged at the terminal, staff is logged off from this merged operation as soon as they clock out. However, they cannot be logged on automatically to the merged operation.

In the mentioned cases, it is recommended to log the operation on and off explicitly, if required.



The "waiting period" and "auto in" options automatically compare the first staff logon within a shift with the rounding rules of HYDRA-PZE if the option "rounding of the first personnel logon" is enabled. This is why, the HYDRA-BDE posting might deviate from the actual time.

If the relevant option is enabled in the basic parameter settings, the personnel logon time in BDE is dated forward to the beginning of the working time temporarily rounded by PZE if a person is logged in by a clocking-in.

Example: a person clocks in at 5.55 a.m. However, the person's working time only starts at 6.00 a.m. according to the shift model and the PZE rounding rules. Waiting period processing logs the person on at 6.00 a.m. in BDE instead at 5.55 a.m. (if enabled at all).

This option only affects such personnel logons in BDE that are performed due to waiting period processing of a person clocking in. This option neither affects staff logging on or off using BDE posting dialogs nor further clocking-in records of the person on the same day.



The waiting period processing is not activated if the person clocks in outside of the skeleton working time, i.e. before the beginning of the skeleton time.



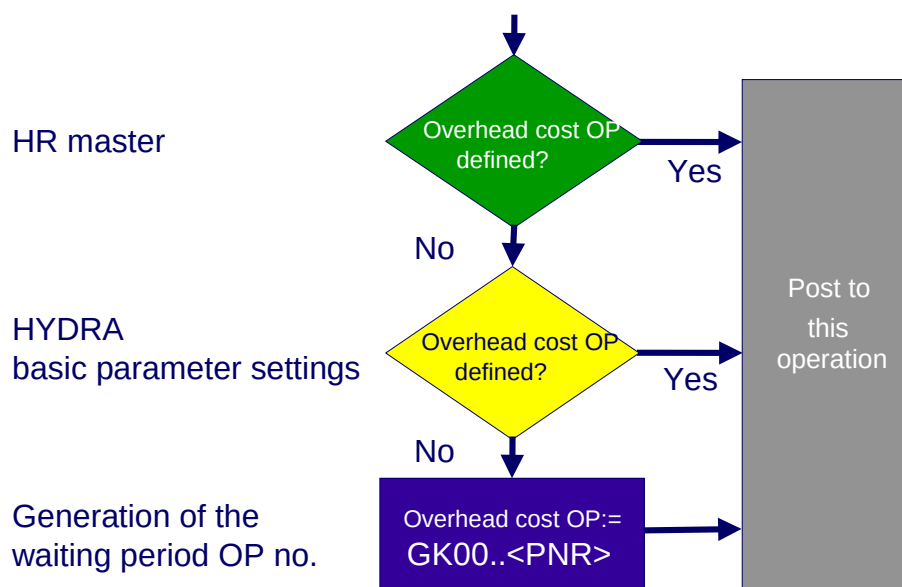
In contrast to HYDRA-BDE, postings in HYDRA-PZE are not transferred ONLINE but at a later point in time. At first they are buffered locally. When it comes to postings that depend on each other (e.g. within the scope of waiting period processing), this might lead to HYDRA-BDE being more up-to-date than HYDRA-PZE postings. In this case, HYDRA-BDE postings are not backdated.

-  Backdating to the beginning of the last shift at most takes place at workplaces/machines, which are directly assigned to a terminal of the operation mode "MDE".
-  When the waiting period function for personnel and machines is used at the same time, the person is postdated to the time when the operation is logged on at most. A waiting period posting for the person is generated for the remaining time, irrespective of whether the waiting period has been exceeded or not. The waiting period for machines always takes priority.
-  If the person is logged off automatically by clocking out, and if quantities in B records are required, he/she has to upload his/her quantities beforehand using partial uploads.

Effects of the settings in the HR master

The recorded waiting period for a person is compared with the person's annual model as it is defined in the [HR master](#); the generated posting of the record type "B" is assigned the ("master") workplace defined in the HR master. If these fields are not filled out the waiting period is 0, provided that the year model is missing or if the workplace is missing it is set to the value 0 or blank.

Finding of the waiting period OP with waiting period processing based on personnel



The resource performance account that is configured in the basic parameter settings is posted, regardless of the object where the waiting period operation is found.

Upload to the PPS system

By default, waiting period postings are not uploaded to the higher-level ERP system. However, if it is required to upload these postings as well, HYDRA can be customized accordingly. To do so, the HYDRA standard interface (HYD-ERP) has to be used.

Status of a waiting period operation

By default, a waiting period operation is always assigned to the status "available (waiting period)", irrespective of whether the waiting period operation has already been posted or if a person or a machine "is waiting".

Processing is not affected, if the status of a waiting period operation or a waiting period order is changed to the "completed" status using the function "change status" in the order overview or order information dialog. The defined waiting period operation is still posted. Please also take into account the notes on changing a waiting period operation in the basic parameter settings.

Reactivation of a waiting period operation

Processing is not affected if a waiting period operation is reactivated using the "reactivate OP" function in the order overview dialog. By default, the status is set to "available (waiting period)" again in HYDRA (this is an internal status with control indicator "U").

Example 1: Effects of the planned working time in HYDRA-PZE on HYDRA-BDE

Prerequisites:

- HYDRA-BDE and HYDRA-PZE are in use
- The option "PZE (in/out) controls BDE" has to be set to "waiting period " or "Auto in".

```

Planned attendance PZE Person 1020
                        6:00 am
                        |_____

Person 1020 clocks in
                        5:55 am
                        |_____

                        6:00 am (determined by the planned attendance)
OP 70010 030 is logged on |_____
Person 1020 is logged on |_____
  
```

Example 2 "waiting period exceeded"

The waiting period is set to 10 minutes in this and the following examples

last logoff (or clock-in) of the person

5:30 am
|-----....

the person is logged on the next time at

6:00 am
|-----

A waiting period posting of 30 minutes is generated for this person and posted on the waiting period RPA for the period of time between 5.30 am and 6.00 am. No backdating takes place.

Example 3 "waiting period not exceeded"

Last logoff or clock-in of the person

5:55 am
|-----....

The person is logged in the next time at

6:00 am
|-----
B| (Backdating to 5:55 am)

The person's logon is backdated to 5.55 am. The five minutes of waiting period are posted on the waiting period RPA. No waiting period posting is generated.

Example 4 "logoff of staff before clocking out in PZE"

The person is logged off

1:55 pm
|-----|

The person clocks out

2:00 pm
|-----|..

In this case, a waiting period posting between the last person that logs off (HYDRA-BDE) and the clock-out (HYDRA-PZE) is generated, even if the waiting period is not exceeded, as there is no (longer) an operation onto which this time could be posted. The five minutes of waiting period are posted onto the RPA defined within the basic parameter settings. The time when the last person logs off is not postdated.

Example 5 – "logon of staff with PZE rounding"

The person arrives at 5.55 am. This time is already dated to 6.00 am by the clocking time gradation function of HYDRA-PZE; i.e. the shop floor collection module (BDE) logs the person on at 6.00 am (HYDRA-BDE does not know the original clocking time in HYDRA-PZE). Consequently, the person works on the waiting period OP as of 6.00 am.

Logging an OP/a person on at 6.05 am triggers a backdating or waiting period posting.

As in this case, the waiting period has not been exceeded (6.00 am - 6.05 am: is a maximum of five minutes, which is less than the 10 minutes waiting period), the logon time for the OP/person is backdated to 6.00 am.

If the OP/person is logged on at 6.10.01 am the waiting period is exceeded and HYDRA generates a waiting period posting of 10 minutes and 1 sec.

The logon time for the OP/person remains at 6.10.01 am.

5 Waiting Period Processing - Machine

Summary

Utilization

Similar to the waiting period processing for staff, this function enables the waiting period processing for machines. Provided that this function is activated, it generates machine-related waiting period postings in the system for times when no operation is logged on.

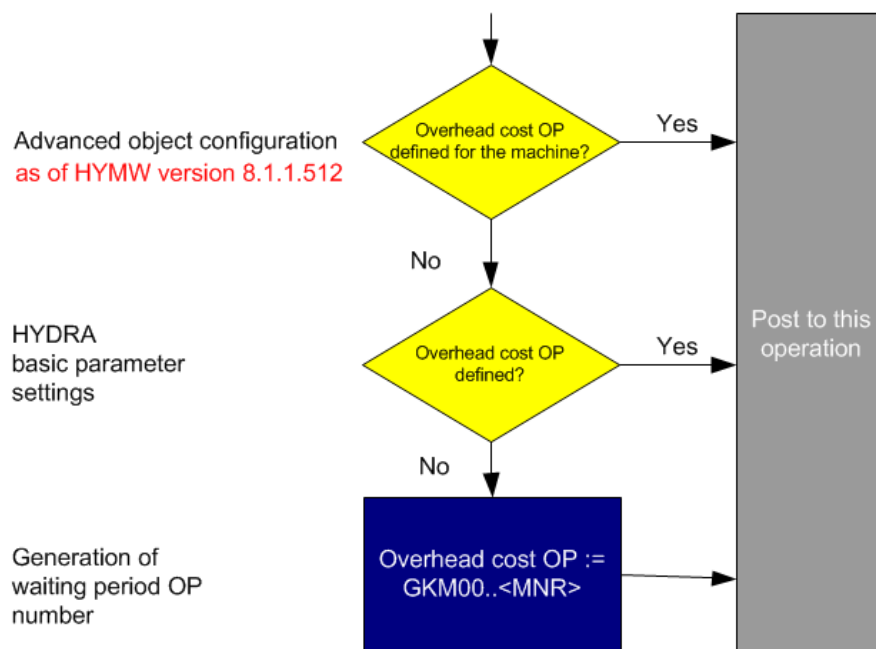
This data represents the unused capacities and, as a result, allows for a statement to be made about the actual utilization of capacities.

Prerequisite

The waiting period processing has been activated in the "[basic parameter settings](#)" → BDE

→ waiting period processing of the machine.

Finding of waiting period OPs with waiting period processing based on machines



Upload to the PPS system

By default, waiting period postings are not uploaded to the higher-level ERP system. However, if it is required to upload these postings as well, HYDRA can be customized accordingly. To do so, the HYDRA standard interface has to be used.

Status of a waiting period operation

By default, a waiting period operation is always assigned to the status "available (waiting period)", irrespective of whether the waiting period operation has already been posted or if a person or a machine "is waiting".

Processing is not affected, if the status of a waiting period operation or a waiting period order is changed to the "completed" status using the function "change status" in the order overview or order information dialog. The defined waiting period operation is still posted. Please also take into account the notes on changing a waiting period operation in the basic parameter settings.

Reactivation of a waiting period operation

Processing is not affected, if a waiting period operation is reactivated using the "reactivate OP" function in the order overview dialog. By default, the status is set to "available (waiting period)" again in HYDRA (this is an internal status with control indicator "U").

6 Activating Waiting Period Processing

Procedure

Workplace for waiting period processing

A special workplace should be set up in an initial step and it should be clear that this is a separate workplace at which waiting periods will be processed.

Setting up a waiting period order

A waiting period order is set up using the [edit orders](#) menu option.

Because waiting period operations are not explicitly posted and are not meant to be confirmed/uploaded to the PPS system in every case, these must be set up

- using order type GKP (to process personal waiting periods)
- using order type GKM (to process machine-related waiting periods).

Initially, the order will be set with the status "available (waiting period)".

Assigning operations to a waiting period order

An operation must now be assigned to the waiting period order. This is done by using the [edit operations](#) menu option. When doing so, make sure that the workplace for waiting period processing that was set up in the initial step is set in the field "workplace".

At this operation, define an authorization level that corresponds to the authorization levels set for the persons listed in the [HR master data](#).

Initially, the operation will be set with the status "available (waiting period)".

The operation is also assigned the processing code "GK".

Optionally, further waiting period operations may be created and assigned using the HR master.



Overhead cost operations may only be changed if they are not active at the moment, i.e. no person or machine may currently be logged on to the OP.

Activating waiting period in the HYDRA basic settings

The waiting period operation that was just set up is then defined in the basic settings (combined order / operation number). The waiting periods are posted to this operation. You cannot assign an operation with a different order type.

Settings in HR master data

What is important for waiting period processing is that for each person for whom postings are carried out by BDE that a year model and a workplace are defined for this person in the [HR master data](#).

As a rule, the authorization level for order postings must be set to > 0 set for a person in the HR master data so that waiting period processing is active for this person.

If during waiting period processing (PZE controls ADE in "Waiting period" mode; "Auto on" or "Out") the operations for the persons are automatically interrupted and in some cases logged back on by PZE Out/In, then these persons must also have all authorization rights to post operations:

- Authorization "Log OP on" must be active.
- Authorization level "OP postings" for the person must be greater than or equal to the authorization level of the operation.

Another option is to define a waiting period OP (order type GKP) to which waiting periods should be posted.